

# Zachary Trenary

Orlando, Florida • 32803

(407) 486-8332 (mobile) • zmichaeltrenary@gmail.com

## Education

**University of Central Florida | Orlando, FL**

B.S. in Computer Science | Fall 2023 – Summer 2026

- Graduated Cum Laude (GPA: 3.82)

## Skills

- **Languages & Frameworks:** Python, C, C++, TypeScript, JavaScript, PHP, SQL/MySQL, HTML/CSS, Flask, React, Next.js
- **Developer Tools & APIs:** Git, Postman, SwaggerHub, Jira, REST API Development, spaCy, OpenCode CLI, Wireshark
- **Cloud & Systems:** AWS (EC2), Linux (Ubuntu), Bash/Shell, LAMP Stack Deployment, Virtualization (Oracle VirtualBox), Networking Fundamentals

## Relevant Coursework

- Computer Science I & II, Senior Design I & II, Discrete Structures I & II, Processes for Object-Oriented Software Development, Systems Software, Computer Logic and Organization, Modeling and Simulation, System Administration and Maintenance, Matrix & Linear Algebra, Statistical Methods I & II

## Projects

**Talking with Tito 3 | Python Flask API + LLM Integration**

01/2026 - 08/2026

*Senior Design Group Project*

- Engineered major v3.0 platform upgrades for an AI language tutor application, enhancing legacy Python Flask and Next.js codebases to improve conversational accuracy and user experience.
- Upgraded backend LLM infrastructure to an optimized Qwen 3.5 model, significantly reducing response latency and expanding context windows for multilingual practice.
- Redesigned front-end UI/UX components in React, introducing real-time star-tracking, dynamic completion meters, multi-chat session management, and administrative console controls.

**LAMP Stack Contact Manager | REST API + MySQL Database**

01/2026 - 02/2026

*POOSD Group Project*

- Engineered a LAMP stack backend featuring a fully documented REST API for user login, registration, and contact management.
- Designed the database architecture, implementing a Users–Contacts schema with enforced foreign key constraints to maintain consistent one-to-many relationships.
- Executed end-to-end testing through Postman and SQL verification, confirming that API endpoints correctly modified the database and met system requirements.

**Analog & Digital Telegraph | Arduino Project + Excel**

04/2023 - 05/2023

*Modeling and Simulation Group Project*

- Led the construction of a physical telegraph system using everyday objects, enabling real-time Morse code communication with a microcontroller and Excel workbook.
- Developed Arduino C++ code to encode outgoing and decode incoming Morse code signals, managing serial communication for seamless data exchange between hardware and software.

## Work Experience

**WDWCP Epcot Parking CM** | Bay Lake, FL

01/2026 - 08/2026

*Disney College Program*

- Managed high-volume parking operations and guest safety in a fast-paced environment, coordinating real-time traffic logistics and maintaining clear communication across live operations.